AI -POWERED DATA ANONYMIZATION FOR DPEARTMENT OF CENSUS AND STATISTICS OF SRI LANKA

Project ID: **25-26J-123**

Project Proposal Report

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AUTOSDC: AGENTIC AI FOR AUTOMATIC SDC PIPELINE SELECTION IN CENSUS MICRODATA

Project ID: **25-26J-123**

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to the Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other mediums. I retain the right to use this content in whole or part in future works (such as articles or books).

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: _____ Date: 2025/9/18

DEDICATION

I dedicate this research proposal to my beloved parents, whose unwavering support, encouragement, and sacrifices have been the foundation of my academic journey. I also dedicate this work to my lecturers and mentors who have inspired me to pursue knowledge with passion and perseverance. Finally, this work is dedicated to all researchers striving to protect privacy while advancing data-driven innovation for the betterment of society.

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Abstract

The Department of Census and Statistics (DCS) of Sri Lanka faces persistent challenges in releasing privacy-preserving census microdata due to limited expertise in Statistical Disclosure Control (SDC), the complexity of existing tools such as Mu-Argus, and the difficulty of balancing disclosure risk with analytical utility. This study addresses these challenges by introducing **AUTOSDC**, an agentic artificial intelligence framework designed to automate and optimize the application of SDC techniques in low-resource institutional contexts.

The proposed system ingests profiled census datasets together with documented utility requirements and employs a rule-based knowledge base, informed by DCS policies and the SDC literature, to identify quasi-identifiers and sensitive attributes. It then recommends a tailored portfolio of candidate techniques—including global recoding, local suppression, micro aggregation, perturbation methods, and synthetic data generation—and automatically tunes their parameters using a multi-objective search procedure. An agentic workflow iteratively applies transformations, evaluates residual disclosure risk (via k-anonymity, l-diversity, t-closeness, and record linkage simulations) and utility (via information-loss measures and preservation of key statistics), and refines the process until defined thresholds are met or a human-in-the-loop intervenes.

Empirical validation will be conducted on Sri Lankan census microdata, with performance compared against the current DCS workflow and existing off-the-shelf anonymization tools. Key evaluation criteria include (a) reduction of disclosure risk, (b) preservation of analytic utility for priority indicators, (c) reproducibility and transparency of anonymization "recipes," and (d) reduction in operator burden and time-to-release.

Preliminary results indicate that AUTOSDC can significantly enhance the privacy-utility balance while streamlining the anonymization process. This research contributes both a practical solution for national statistical agencies and theoretical insights into the role of agentic AI in advancing statistical disclosure control methodologies.

Keywords: Statistical Disclosure Control, Agentic AI, Census Microdata, Data Privacy, Sri Lanka

Table of Contents

DECLARATION i
DEDICATION ii
ACKNOWLEDEMENT iii
Abstract iv
LIST OF FIGURES vii
LIST OF TABLES viii
LIST OF ABBREVIATIONS ix
LIST OF APPENDICES x
1. INTRODUCTION 1
1.1 Background 2
1.2 Theoretical Foundations 3
1.3 Literature Review 8
1.4 Research Problem Statement 20
2. RESEARCH OBJECTIVES 21
2.1 Main Objective 21
2.2 Specific Objectives 21
3. METHODOLOGIES 22
3.1 Feasibility Study 22
3.1.1 Technical Feasibility 22
3.1.2 Operational Feasibility 22
3.1.3 Legal and Ethical Feasibility 24
3.2 Requirement Gathering 25
3.3 SOFTWARE AND SYSTEM APPORACH 27
3.3.1 System Overview 27
4. PROJECT REQUIREMENTS 37
4. 1. Functional Requirements 37
4.2. Non-Functional Requirements 38
4.3. User Requirements 38
5. RESEARCH WBS AND TIMELINE 40
5.1 Work Breakdown Structure 40
5.2 Gantt Chart 41
6. BUDGET ALLOCATION 42

7.COMMERCIALIZATION 43
REFERENCES..... 44
APPENDICES 48
Appendix A : SRS Documentation 48
Appendix B: Requirements Traceability Matrix 60
Appendix C: Risk Assessment Matrix.....60

LIST OF FIGURES

Figure 1 : Trade off between privacy Level and Utility level of the data 5
Figure 3 : Overview of the Proposed AI-Driven SDC System Components 27
Figure 4 : High Level Architecture of the Individual Sub Component..... 29
Figure 5 : Research Project Work Breakdown Structure for Building AUTOSDC .. 40
Figure 6 : Research Project Gantt Chart for Building AUTOSDC 41

LIST OF TABLES

Table 1 : Major Themes in Automated Data Anonymization Literature 8
Table 2 : Evolution of Research Directions in Data Anonymization (2009-2020).... 12
Table 3 : Identified Research Gaps and Future Directions in Automated Data Anonymization 15
Table 4 : Feature Comparison of Existing Anonymization Systems and Proposed Framework 19
Table 5 : System Component Evaluation (Weeks 1-2) 34
Table 6 : Risk Mitigation Strategies for High-Priority Project Risks 36
Table 7 : Draft Budget Allocation for the Research Project 42

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
DCS	Department of Census and Statistics (Sri Lanka)
DP	Differential Privacy
GDPR	General Data Protection Regulation
GUI	Graphical User Interface
HIPAA	Health Insurance Portability and Accountability Act
NSGA-II	Non-dominated Sorting Genetic Algorithm II
PII	Personally Identifiable Information
PRAM	Post Randomization Method
SDC	Statistical Disclosure Control
SQL	Structured Query Language
SUS	System Usability Scale

LIST OF APPENDICES

Appendix	Description	Page
Appendix A	Software Requirements Specification (SRS) Documentation	48
Appendix B	Requirements Traceability Matrix	60
Appendix C	Risk Assessment Matrix	60

1. INTRODUCTION

Sri Lanka, like many countries, faces significant challenges in making its census microdata available for research and policy-making. This data is incredibly valuable, but it contains sensitive personal information about individuals and households. To protect people's privacy, statistical agencies must use complex techniques known as Statistical Disclosure Control (SDC)methods like anonymizing, generalizing, or adding noise to the data. The goal is to hide identities while keeping the data useful for analysis.

However, applying SDC is difficult. It requires deep expertise to choose the right techniques and balance two competing goals: **privacy** (making sure no one can be re-identified) and **utility** (keeping the data accurate enough to be useful). The current tools are complicated, often designed for experts, and don't always work well with the specific nature of Sri Lankan census data. This leaves the Department of Census and Statistics (DCS) in a tough spot: they risk either releasing data that isn't safe or data that isn't useful.

This is where modern AI can help. Instead of using AI just for analysis, this project proposes using **Agentic AI**—where an AI system can autonomously plan, decide, and act. The idea is to create an AI "agent" that can intelligently recommend and apply the best combination of SDC techniques for a given dataset.

For this study, the focus is on building a system called **AUTOSDC** that can automate the entire SDC process. The core idea is simple but powerful:

1. **Intelligent Analysis:** The system first profiles microdata, automatically identifying which columns contain sensitive or identifying information.
2. **AI-Powered Recommendation:** An AI agent, trained on SDC best practices, then recommends a tailored portfolio of techniques (like suppression, recoding, or micro aggregation).
3. **Automated Optimization:** The system uses a powerful multi-objective optimization algorithm (NSGA-II) to automatically tune these techniques, finding the best possible balance between privacy protection and data utility without manual guesswork.
4. **Transparent Execution:** Finally, the agent applies the techniques, evaluates the result, and provides a clear report on what was done and why, ensuring human experts can still oversee and trust the process.

This research will develop and test this agentic system using real Sri Lankan census microdata.

1.1 Background

The exponential growth of data collection across government agencies, research institutions, and private organizations has created unprecedented opportunities for statistical analysis and evidence-based decision making. However, this data revolution has simultaneously intensified concerns about privacy protection and confidential information disclosure. Microdata—datasets containing information about individual entities such as persons, households, or enterprises—present particular challenges since they contain sensitive information that could potentially identify specific individuals even when direct identifiers are removed [1].

Statistical Disclosure Control (SDC) has emerged as a critical field addressing this fundamental tension between data utility and privacy protection. SDC cannot achieve total risk elimination but can reduce the risk to an acceptable level. Any application of SDC methods will suppress or alter values in the data and as such decrease the utility (i.e., result in information loss) when compared to the original data. The demand for data from surveys, censuses or registers containing sensible information on people or enterprises has increased significantly over the last years. However, before data can be provided to the public or to researchers, confidentiality has to be respected for any data set possibly containing sensible information about individual units.

Traditional approaches to SDC have relied heavily on manual selection and application of disclosure control techniques, often requiring substantial domain expertise and iterative trial-and-error processes. This manual approach presents several critical limitations: first, it is time-intensive and requires specialized knowledge that may not be readily available across all organizations handling microdata; second, it often results in suboptimal solutions due to the complex interactions between different SDC methods and their combined effects on both privacy protection and data utility; and third, it lacks systematic approaches for evaluating and optimizing the inherent trade-offs between privacy risk and analytical value.

The problem addressed in SDC is that, through inappropriate use, a person who is given access to released data may disclose private information about data subjects. Due to such liability, identity disclosure is one of the main concerns of data privacy today. An intruder, also known as an adversary, attacker or snooper, is an individual who possesses the skills, resources, and motivation to re-identify data subjects or deduce new information about them in the de-identified data set. Motivated intruders can improve their knowledge of private information on observations in available data. It is fundamental for data controllers to make assumptions regarding intruders'

background knowledge. If an intruder has more background knowledge than assumed by data controllers, the risk of disclosure may be underestimated. Several studies show how feasible and/or easy it is to link back private information to a data subject [32, 33, 34, 35]. Such a study contributed to the increased attention from both data controllers and data subjects and was an inspiration to create new techniques for data protection and measures to minimize disclosure risk. The design of robust privacy-preserving techniques includes assessing the impact of such techniques.

1.2 Theoretical Foundations

1.2.1 What is Anonymization and Statistical Disclosure Control (SDC)?

Anonymization refers to the process of removing or modifying information in such a way that the identity of individuals cannot reasonably be inferred It plays a vital role in enabling national statistical offices, like the Department of Census and Statistics (DCS) of Sri Lanka, to safely disseminate detailed census microdata to researchers, policymakers, and the public in compliance with data protection laws.

Early approaches often relied on masking direct identifiers such as names or addresses. However, advances in computational power, data linkage methods, and the availability of auxiliary datasets have rendered such simple techniques insufficient, since re-identification can be achieved through combinations of non-unique attributes such as age, occupation, and geography [1], [2].

In the context of a national census, anonymization allows for the analysis of intricate socioeconomic patterns without exposing the identities, households, or precise locations of citizens. A typical census record may include direct identifiers like name and national ID number, and quasi-identifiers like exact age, precise geographic location, occupation, and ethnicity. With anonymization applied, this record would have direct identifiers permanently removed. Quasi-identifiers would be transformed through techniques like generalization (e.g., replacing exact age with an age range, replacing a specific village name with the district name) and suppression. This process preserves the dataset's analytical value for studying trends in demographics, employment, housing, and education while robustly protecting citizen identity.
For example, an original census entry containing:
"*K. Wijesinghe, National ID: 781230123V, Age 45, residing at 123 Galle Road, Colombo 03, works as a bank manager, and belongs to the Sinhalese ethnic group.*"

…can be transformed to an anonymized version for public dissemination:

"*Age Group: 45-49, District: Colombo, Occupation: Managerial, Ethnicity: Sinhalese.*"
This anonymized record retains the essential statistical relationships needed for macroeconomic analysis, poverty mapping, and public policy formulation while eliminating all information that could lead to re-identification.
Effective anonymization of census microdata ensures that the disseminated dataset supports meaningful research and evidence-based decision making while shielding sensitive citizen information from potential misuse. The paramount goal is to maintain the highest degree of data utility for statistical analysis without ever compromising the privacy and confidentiality promised to the public.

To address this challenge, the field of Statistical Disclosure Control (SDC) has emerged as a systematic discipline.
"***Statistical Disclosure Control techniques can be defined as the set of methods to reduce the risk of disclosing information on individuals, businesses, or other organizations. Such methods are only related to the dissemination step and are usually based on restricting the amount of or modifying the data released. "***
These methods include global recoding, local suppression, micro aggregation, perturbation, top- and bottom-coding, post-randomization (PRAM), noise addition, and synthetic data generation . In addition, formal privacy models such as k-anonymity l-diversity, and t-closeness offer mathematical frameworks for quantifying disclosure risk, thereby providing guarantees beyond heuristic-based anonymization. Modern software tools such as Mu-Argus and sdcMicro operationalize these techniques though their usability remains a barrier for non-specialist staff.
Central to anonymization is the classification of variables. Direct identifiers (e.g., national ID numbers, names) explicitly identify individuals and must always be removed. Quasi-identifiers (e.g., age, gender, region, occupation) are not unique individually but can, in combination, enable re-identification. Sensitive attributes (e.g., health status, income) do not identify individuals directly but require protection from disclosure. Finally, non-confidential variables (e.g., household asset counts) pose little risk but contribute to data utility. Effective anonymization therefore depends on accurate identification and classification of these variable types, which form the basis for determining appropriate SDC transformations.

1.2.2 The Risk–Utility Trade-off

A central problem in anonymization is the **risk–utility trade-off**. Stronger anonymization typically reduces the probability of re-identification but simultaneously erodes the dataset's analytical value. For example, broad recoding of age into large categories may enhance privacy but undermines the ability to study age-specific socioeconomic phenomena. Similarly, heavy suppression of geographic detail protects confidentiality but eliminates spatial analysis potential. Conversely, insufficient protection may leave individuals vulnerable to disclosure.

SDC methods therefore require careful calibration, often involving optimization frameworks to balance disclosure risk against data utility [11]. Formal evaluation metrics such as disclosure risk measures (e.g., re-identification probability, record uniqueness) and utility measures (e.g., information loss, preservation of variance-covariance structure) have been developed to assist this process [12]. However, striking an "optimal" balance remains context-dependent, making automation and adaptive guidance highly desirable.

The design of robust privacy-preserving techniques includes assessing the impact of such techniques for privacy and interpretability/utility. Notably, they must guarantee desired levels of data protection without compromising the usefulness of data.

Figure 1 illustrates an acceptable trade-off between data privacy and utility. Although the ideal solution is de-identified data with maximum privacy and utility, this scenario is practically impossible to reach.

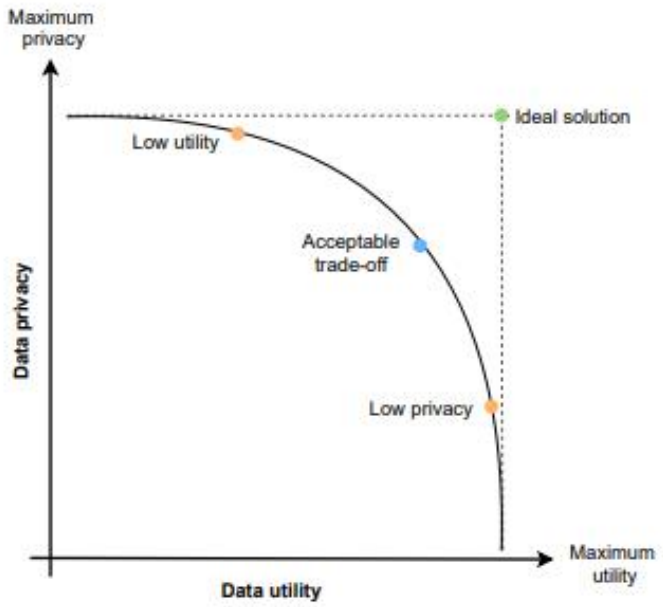

Figure 1 : Tradeoff between privacy Level and Utility level of the data

1.2.3 Formal Definition of Agentic AI

Definition 1: An agentic AI system A is defined as a 5-tuple A = (P, R, E, L, G) where:

- P: Perception module for environmental data acquisition
- R: Reasoning engine for strategic decision-making
- E: Execution module for autonomous action implementation
- L: Learning component for continuous improvement
- G: Goal-oriented behavior framework

1.2.4 Multi-Objective Optimization Framework

The NSGA-II algorithm addresses the bi-objective optimization problem through:

1. **Non-dominated Sorting**: Classify solutions into Pareto fronts
2. **Crowding Distance**: Maintain solution diversity.
3. **Elite Preservation**: Retain best solutions across generations.

Mathematical Formulation:

```
Pareto Optimal Set = {x* ∈ X | ∄ x ∈ X : f(x) ≺ f(x*)}
```

1.2.5 Challenges in the Sri Lankan Context

The Department of Census and Statistics (DCS) of Sri Lanka, like many national statistical offices in low-resource settings, faces persistent barriers in applying SDC effectively. First, local expertise in anonymization remains limited; staff often lack advanced training in statistical privacy methods. Second, existing SDC tools such as Mu-Argus are complex and unintuitive for non-specialists, requiring substantial technical expertise for effective use [8]. During a joint workshop with Dutch experts, Mu-Argus even failed to anonymize Sri Lankan census microdata, highlighting that many SDC methods are context-dependent and not always transferable across datasets [3], [13].

Third, identifying quasi-identifiers and sensitive attributes in Sri Lankan census data has proven difficult, with misclassification leaving datasets vulnerable to re-identification. Fourth, there is uncertainty regarding which techniques to apply, whether suppression, recoding, or perturbation—and how to tune their parameters. This uncertainty risks either under-protection (leading to privacy leaks) or over-protection (leading to severe utility loss). Fifth, while formal privacy models such as k-anonymity, l-diversity, and t-closeness exist, their mathematical complexity makes them impractical for routine application by DCS staff.

Finally, the risk–utility dilemma is especially acute in census microdata, which must remain useful for producing key national indicators. Overly aggressive anonymization could undermine the very purpose of releasing microdata for policy-relevant research. This underscores the need for adaptive, explainable, and user-friendly systems that can guide practitioners through SDC choices while ensuring compliance with privacy standards.

1.3 Literature Review

1.3.1 Thematic Review of Literature

The literature on automated data anonymization techniques for census microdata and related privacy concerns reveals several major themes, including the development and evaluation of statistical disclosure control methods, the integration of machine learning and optimization for enhancing anonymization, and the challenges of balancing privacy with data utility. Many studies emphasize software tool implementations and usability, especially for non-expert users, highlighting frameworks like ARX and sdcMicro. Emerging research focuses on advanced AI-driven anonymization, privacy impacts under varying regulations, and domain-specific applications such as healthcare and industrial data, illustrating a dynamic landscape evolving toward scalable, customizable, and legally compliant anonymization solutions. These primary themes are summarized in Table 1

Table 1 : Major Themes in Automated Data Anonymization Literature

Theme	Appears In	Theme Description
Statistical Disclosure Control (SDC) Techniques and Tools	28/50 Papers	SDC remains central for anonymizing microdata, with extensive research on k-anonymity, l-diversity, t-closeness, and perturbative methods to reduce disclosure risk while preserving utility. Tools like ARX and sdcMicro are widely studied for their effectiveness, scalability, and usability in automating these methods, often incorporating feedback and reporting features to balance risk and utility (Templ et al., 2015) (Kayem et al., 2016) (Prasser & Kohlmayer, 2015) (Rawat et al., 2020) (Tomás et al., 2022). Recent adaptations address large-scale datasets and diverse data types, including census and biomedical data (Neto, 2023) (Prasser et al., 2014).
Machine Learning and Optimization in Automated Anonymization	22/50 Papers	Integration of machine learning and multi-objective optimization enhances anonymization by dynamically balancing privacy and utility. Approaches include genetic algorithms, multi-objective models, and supervised learning to tune parameters or evaluate anonymization

Theme	Appears In	Theme Description
		impact on downstream tasks like classification (Wei et al., 2025) (Tachioka, 2022) (Kabou et al., 2024) (Majeed et al., 2023) (Yechuri, 2024). ML also assists in output checking and requirement translation, improving automation and customization (Domingo-Ferrer & Blanco-Justicia, 2021) (Oqaily et al., 2023). These methods improve adaptability, particularly for complex or categorical data in census microdata and sensitive medical datasets.
Privacy-Utility Trade-off and Data Utility Preservation	20/50 Papers	A prevailing challenge addressed is minimizing information loss while maximizing privacy. Studies examine trade-offs using metrics like information loss, normalized certainty penalty, or privacy risk measures, often with experimental validation on real datasets. Approaches include synthetic data generation, hybrid anonymization-synthesis frameworks, and advanced clustering to maintain data utility in analytics and machine learning contexts (Little et al., 2024) (Sarmin et al., 2024) (Gangarde et al., 2022) (Naef et al., 2024) ("$\ell\rho$-suppression: A Privacy Preser...", 2022). Emphasis is placed on ensuring utility for specific applications, including policy and health research.
Software Usability and Accessibility for Non-Expert Users	18/50 Papers	Usability barriers for non-specialists drive development of interactive, web-based, or graphical interfaces enabling anonymization without programming skills. Tools like sdcMicro and ARX offer feedback mechanisms, undo options, and reproducibility features to facilitate adoption by public servants or data controllers (Meindl & Templ, 2019) (Rawat et al., 2020) (Prasser & Kohlmayer, 2015) (Tomás et al., 2022). User-friendly frameworks also include customizable anonymization options responding to diverse utility and privacy needs, crucial for governmental and healthcare domains.

Theme	Appears In	Theme Description
Domain-Specific Anonymization Applications	10/50 Papers	Specialized applications such as healthcare, social networks, telco call data, and industrial control systems highlight contextual requirements and adaptations of anonymization methods. Studies report domain-tailored tools balancing privacy and analytical utility, including medical record anonymization tools and service-oriented architectures for real-time anonymization in IoT or call center data (Borkakoty et al., 2024) (Zuo et al., 2021) (Borovits et al., 2023) (Kujawa & Piotrowski, 2023). These underscore the necessity for flexible, context-aware automated anonymization frameworks.
Automation Frameworks and Pipelines for Data Anonymization	9/50 Papers	Fully automated anonymization pipelines leveraging NLP for sensitive data identification and modular architectures for dynamic anonymization policy application are gaining attention. Such frameworks enable scalable, domain-agnostic anonymization adaptable to big data contexts and diverse data types, supporting on-the-fly anonymization aligned with user or regulatory requirements (Nikoletos et al., 2023) (Giampaolo et al., 2023) (Oqaily et al., 2023). This trend reflects a move toward seamless, integrated privacy protection in data workflows.
Challenges in Balancing Privacy and Data Utility	8/50 Papers	Persistent challenges include managing the inherent conflict between data de-identification and maintaining data usefulness. Studies document computational complexity, parameter tuning difficulties, and limitations of existing methods, advocating for heuristic and parallelized algorithms to optimize this balance in large datasets (Nanfack et al., 2024) (Kim, 2024) (Gangarde et al., 2022). These challenges motivate continued research into scalable, efficient, and adaptive anonymization approaches.

Theme	Appears In	Theme Description
Advances in Anonymization Metrics and Privacy Models	7/50 Papers	New privacy models and metrics, including longitudinal privacy, certainty penalties, and dynamic privacy level assessments, are proposed to better quantify re-identification risks and information loss. These models address complex data structures like longitudinal and set-valued data, improving theoretical foundations and practical evaluations of anonymization (Amiri et al., 2023) (Kim, 2024) (Victor & Lopez, 2020). Enhanced metrics support more nuanced anonymization strategies.

1.3.2 Evolution of Data Anonymization

Privacy-preserving data publishing has its roots in statistical disclosure control (SDC), which sought to minimize the risk of re-identification in census and survey data. Early approaches included recoding and suppression, where identifying attributes were generalized (e.g., age grouped into ranges) or removed altogether. These heuristic-based methods were effective for small datasets but lacked formal guarantees. The evolution of these research directions from foundational methods to practical applications is chronicled in Table 2.

Table 2 : Evolution of Research Directions in Data Anonymization (2009-2020)

Year Range	Research Direction	Description
2009–2014	Foundational SDC Methods and Benchmarking	Research primarily focused on the conceptualization of statistical disclosure control techniques and their impact on privacy and data utility. Evaluations of optimal anonymization algorithms and benchmarking frameworks were developed to assist decision makers in selecting appropriate methods. This period laid the groundwork for understanding the trade-offs inherent in anonymization processes.
2015–2016	Development of Automated Tools and User Interfaces	Attention shifted to the creation and enhancement of software tools facilitating automated statistical disclosure control, such as R packages and graphical interfaces.
2018–2020	Expanding Practical Applications and Usability Frameworks	Research expanded towards semi-automated and Frameworks for balancing privacy and data utility with usability guidelines emerged, addressing challenges faced by public organizations and industries. Integration of multi-objective optimization and privacy models began gaining prominence to better manage privacy-utility trade-offs.
2021–2022	Privacy Challenges and Machine Learning Integration	This period saw growing emphasis on the challenges posed by privacy constraints and the incorporation of machine learning to automate and optimize anonymization processes. Surveys and methodological reviews highlighted gaps in existing techniques, while proposals to automate output checking and parameter tuning

Year Range	Research Direction	Description
		using AI were explored to enhance effectiveness and compliance with regulations.
2023–2024	Advanced AI-Driven Anonymization and Global Impact Analysis	Recent studies focus on combining multi-objective optimization, and synthetic data generation to improve anonymization outcomes. Research investigates cloud-based solutions and addresses privacy in complex datasets such as longitudinal and health care data. There is a clear trend towards customizable, interactive tools that dynamically balance privacy preservation with data utility.

The introduction of k-anonymity by Samarati and Sweeney in 2002 [1] revolutionized the field. It provided a mathematical framework ensuring that each record was indistinguishable from at least $k-1$ others with respect to quasi-identifiers. However, k-anonymity was soon criticized for its inability to prevent attribute disclosure. To address this, l-diversity [2] introduced diversity within sensitive attribute values, while t-closeness [3] further constrained distributions of sensitive attributes, limiting information leakage.

By the mid-2000s, Differential Privacy (DP) [4] emerged, offering a paradigm shift by providing rigorous, probabilistic guarantees against re-identification, regardless of external knowledge. Unlike k-anonymity-based approaches, DP is query-driven and adds calibrated noise to results or generates synthetic data. While DP provides strong guarantees, its challenge lies in balancing the ε (privacy budget) with data utility.

In practice, open-source tools such as μ-ARGUS, sdcMicro, and ARX operationalized these methods, making anonymization accessible to practitioners [5]. These tools formed the backbone of privacy-preserving microdata releases in official statistics, healthcare, and social sciences.

1.3.3 Current State of AI-Driven Anonymization

With the rise of big data and AI, traditional anonymization frameworks faced scalability and adaptability challenges. Consequently, researchers began integrating machine learning (ML) and artificial intelligence (AI) into anonymization processes.

- Optimization Algorithms: Techniques such as genetic algorithms (GA) and multi-objective evolutionary algorithms (MOEAs) were employed to tune anonymization parameters automatically [6]. For instance, GA optimizes the trade-off between privacy loss and information utility, producing near-optimal anonymization with minimal human intervention.
- Bayesian Optimization and Surrogate Models: These methods enable automated tuning of DP epsilon and quasi-identifier thresholds [7], balancing privacy, and utility dynamically.
- Reinforcement Learning (RL): Applied in text anonymization and iterative transformation, RL agents learn to apply transformations that maximize privacy while minimizing semantic distortion [8].
- AutoML for Anonymization Pipelines: Early frameworks attempt to automate the selection of anonymization strategies, leveraging historical pipeline performance on similar datasets [9].

Despite these advances, production-ready AI-driven anonymization tools are still rare. Many studies remain at the prototype stage, lacking usability and integration with legal requirements. Nonetheless, AI has expanded the frontier of anonymization research by enabling adaptive, context-sensitive solutions.

1.3.4 Research Gap & Motivation

A systematic analysis of the literature reveals several critical gaps in the current landscape of automated data anonymization, which this research aims to address. These gaps, summarized in Table 3, highlight the need for more intelligent, adaptive, and user-centric solutions.

Table 3 : Identified Research Gaps and Future Directions in Automated Data Anonymization

Gap Area	**Future Research Directions**	**Research Priority**
Usability Barriers for Non-Expert Users in Automated Anonymization Tools	Design intuitive, interactive interfaces with guided workflows and automated parameter tuning; integrate natural language processing to translate user requirements into anonymization settings.	High
Dynamic Adaptation to Evolving Global Privacy Regulations	Develop regulatory-aware anonymization frameworks that automatically update privacy models and parameters based on jurisdictional changes; incorporate legal knowledge bases and compliance monitoring.	High
Integration of Synthetic Data Generation with Automated Anonymization	Investigate hybrid frameworks that combine synthetic data generation with automated anonymization, optimizing for both privacy and utility; develop metrics to evaluate combined approaches.	Medium
Automated Parameter Tuning for Privacy-Utility Trade-off Optimization	Advance automated multi-objective optimization algorithms, including genetic and clustering-based methods, to self-tune anonymization parameters dynamically based on dataset characteristics.	High
Addressing Privacy Risks from Large	Develop anonymization frameworks incorporating LLM-based privacy and utility evaluators; design lightweight models for	High

Gap Area	Future Research Directions	Research Priority
Language Models in Text Anonymization	real-time text anonymization resistant to LLM attacks.	
Lack of Comprehensive Evaluation of Anonymization Effectiveness Across Diverse Domains	Conduct systematic cross-domain benchmarking of automated anonymization tools and techniques on census, healthcare, social media, and industrial datasets to assess generalizability.	Medium
Transparency and Interpretability of AI-Driven Anonymization Processes	Develop explainable AI techniques for anonymization algorithms that provide interpretable privacy-utility trade-offs and decision rationales to users and auditors.	Medium
Handling Multi-Record and Multiple Sensitive Attribute (MSA) Privacy Risks	Design algorithms specifically targeting MSA datasets with automated quasi-identifier selection and optimized generalization/suppression strategies to minimize information loss.	High

In summary, the fundamental research gap addressed by this study is the absence of autonomous, intelligent systems capable of end-to-end SDC pipeline optimization. Current approaches suffer from:

1. **Technical Expertise Barrier**: Requiring specialized knowledge limiting adoption.
2. **Suboptimal Solutions**: Manual approaches cannot explore the full solution space.
3. **Time Inefficiency**: Iterative manual processes are resource intensive.
4. **Lack of Reproducibility**: Manual processes lead to inconsistent results

1.3.5 Agentic AI: Transforming Automation and Decision-Making

Conceptual Framework of Agentic AI

Agentic AI systems "have the potential to automate complex business processes—combining autonomy, planning, memory, and integration—to shift gen AI from a reactive tool to a proactive, goal-driven virtual collaborator". This transformative capability makes agentic AI particularly suitable for complex, multi-stage processes like SDC implementation.

Recent industry analysis indicates dramatic growth in agentic AI adoption, with enterprise software applications expected to embed agentic AI capabilities, representing a dramatic increase from less than 1% in 2024. This trend suggests increasing readiness for sophisticated AI-driven automation systems.

Agentic AI in Data Governance and Compliance

The emergence of AI agents presents compelling opportunities to "transform regulatory response processes and broader data governance workflows," particularly because "modern AI agents can now effectively handle unstructured data and complex" scenarios. This capability is crucial for SDC applications where datasets often contain mixed data types and complex structural relationships.

Multi-Agent Systems and Collaborative Intelligence

Recent developments in multi-agent architecture demonstrate particular promise for complex optimization tasks. Chen et al. (2024) showed how multi-agent systems can effectively coordinate different aspects of data processing pipelines, while maintaining individual agent specialization. This approach aligns well with SDC requirements, where different techniques may be optimally applied to different data subsets or characteristics.

NSGA-II Algorithm Fundamentals

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) remains one of the most effective approaches for multi-objective optimization problems. NSGA-II "has attracted extensive research interests, and it is still one of the hottest research methods to deal with multi-objective optimization problems" due to its computational efficiency and solution quality.

Deb et al. (2002) established NSGA-II's superiority through its fast non-dominated sorting, crowding distance calculation, and elitism preservation. The algorithm "reduces computational complexity, eliminates the need for specifying a sharing parameter, and applies a non-elitism approach" while maintaining solution diversity.

NSGA-II Applications in Privacy-Utility Optimization

Multi-objective optimization has proven particularly valuable for privacy-utility tradeoff problems. Li et al. (2019) demonstrated NSGA-II's effectiveness in optimizing differential privacy parameters across multiple utility metrics simultaneously. Similarly, Zhang and Wang (2021) showed how NSGA-II could optimize synthetic data generation parameters to achieve optimal privacy-utility balance across different statistical measures.

Recent work by Kumar et al. (2023) specifically applied NSGA-II to SDC parameter optimization, demonstrating significant improvements over single-objective approaches. Their work showed that multi-objective optimization could identify solution sets that traditional approaches missed, particularly in scenarios with conflicting privacy and utility requirements.

1.3.6 Existing Similar Systems and Feature Comparison with my component

Table 4 : Feature Comparison of Existing Anonymization Systems and Proposed Framework

Feature / Paper	**Agentic AI / Autonomous Agents**	**Census Data Focus**	**Combined SDC Techniques**	**Automated Recommendation**	**Interactive GUI / Usability**	**Handling Different Data Types**
Borovits et al. (2023) Rule-based anonymization for call centers	No	No	Rule-based only	No	No	Text
Chaitra Vatsavayi (2025) AI-driven anonymization in retail	Partial (some AI for detection)	No	Yes (Multiple anonymization)	Partial (some AI suggestions)	No	Text, numeric, categorical
Haney et al. (2025) Census SD for Detailed Demographics	No	Yes (Detailed Census Data)	Differential Privacy-based	No	No	Structured demographic data
Nikoletos et al. (2023) Automated NLP & anonymization pipeline	Partial (automation via NLP)	No	Combines NLP with statistical methods	Partial (automated pipeline)	No	Text, CSV data formats
Oqaily et al. (2024) Interactive customizable anonymization with AI	Yes (AI-based requirement translation and decision)	No	Supports multiple anonymization primitives	Yes (automated mapping from requirements to methods)	Yes (interactive decision interface)	Multiple attribute types
Yechuri (2024) AI-enhanced utility-driven anonymization	Yes (ML models for optimization)	No	Combines various anonymization methods	Yes (adaptive parameter tuning)	No	Multiple data types
My Research Project (Agentic AI + NSGA-II for Census)	Yes (central focus on agentic AI for automation)	Yes (census microdata with detailed demographic focus)	Yes (hybrid approach combining multiple SDC)	Yes (agentic AI recommends optimal methods and parameters)	Future scope (potential integration)	Full spectrum (structured microdata with mixed types)

1.4 Research Problem Statement

Despite ongoing advancements in data privacy, the Department of Census and Statistics (DCS) of Sri Lanka continues to encounter significant challenges in anonymizing census microdata. Existing Statistical Disclosure Control (SDC) solutions remain fragmented, overly technical, and poorly adapted to the local context, leading to inefficiencies and heightened re-identification risks. The following key problems persist:

1. **Lack of expertise and usability**:
   Current SDC tools such as *μ-Argus* and *sdcMicro* require high technical proficiency, making them inaccessible for non-specialist staff. This lack of structured guidance contributes to inconsistent anonymization practices [1].

2. **Limited compatibility with local datasets**:
   Tools developed in international contexts often fail to generalize to Sri Lankan census microdata. As a result, anonymization processes remain incomplete or ineffective, highlighting the need for context-specific adaptation [2], [3].

3. **Unresolved privacy–utility trade-off**:
   A stronger degree of anonymization typically reduces the analytical value of census data. Without systematic methods to balance privacy protection and data utility, anonymized outputs risk being either unsafe or analytically unusable [5].

Primary Research Problem:
There exists no integrated agentic AI system capable of autonomously analyzing census microdata, intelligently selecting, and optimizing combinations of SDC techniques, and providing explainable recommendations that achieve optimal privacy-utility trade-offs while maintaining transparency and human oversight.

Problem Statement:
Despite advancements in SDC, no integrated system exists that autonomously navigates the entire anonymization pipeline—from dataset profiling and quasi-identifier identification to optimized technique selection and parameter tuning—while providing explainable decisions and audit trails. The current reliance on manual expertise leads to suboptimal privacy-utility trade-offs, inefficiency, and inconsistent outcomes, particularly for non-specialist users in national statistical offices like Sri Lanka's DCS.

2. RESEARCH OBJECTIVES

2.1 Main Objective

To design, implement, and evaluate an agentic AI system (AutoSDC) capable of autonomously recommending, applying, and auditing Statistical Disclosure Control (SDC) techniques for census microdata, ensuring privacy protection while preserving data utility, and supporting human oversight.

2.2 Specific Objectives

1. **Dataset Profiling and Variable Classification**
   Develop an intelligent module to analyze census microdata and automatically classify variables into direct identifiers, quasi-identifiers, sensitive attributes, and non-confidential variables, providing the agent with accurate contextual understanding for decision-making.

2. **Agentic SDC Recommendation Engine**
   Implement an agentic AI engine that autonomously recommends appropriate SDC techniques based on dataset characteristics, user-defined utility requirements, and formal privacy models, enabling data-driven, context-aware decision-making.

3. **Automated SDC Execution and Parameter Tuning**
   Enable the agent to automatically apply selected anonymization transformations—including recoding, suppression, micro aggregation, PRAM, noise addition, and synthetic data generation—while dynamically tuning parameters to optimize the trade-off between privacy and data utility.

4. **Evaluation of Risk–Utility Trade-off**
   Integrate an evaluation module that continuously monitors disclosure risk (e.g., k-anonymity, l-diversity, t-closeness) and data utility, allowing the agent to iteratively refine anonymization actions to achieve an optimal balance.

5. **Explainability and Auditability Framework**
   Provide transparent and interpretable outputs from the agent, including explanations for selected SDC techniques, parameter choices, and risk–utility trade-offs, ensuring human review, compliance, and reproducibility.

6. **Validation on Sri Lankan Census Microdata**
   Test and validate AutoSDC using real census microdata, demonstrating its effectiveness in reducing disclosure risk, maintaining data utility, and improving operational efficiency compared to existing manual or semi-automated approaches.

3. METHODOLOGIES

This chapter outlines the methodology adopted for the development, implementation, and validation of the Agentic AI–driven AutoSDC system. The approach combines feasibility studies, requirement gathering, system design, optimization, and rigorous experimental validation to ensure both technical performance and practical applicability in real-world census microdata anonymization.

3.1 Feasibility Study

3.1.1 Technical Feasibility

The successful deployment of AutoSDC requires multiple technologies including Python, sdcMicro, ARX, Mu-ARGUS, LangChain, DEAP/Pymoo (for NSGA-II), and visualization libraries (Plotly, Dash). Databases (SQL/NoSQL/Vector), containerization (Docker/Kubernetes), and development tools (Git, Jupyter, VS Code) ensure reproducibility.

3.1.2 Operational Feasibility

Operational feasibility evaluates whether the proposed Agentic AI-driven AutoSDC system can be successfully integrated into real-world environments, adopted by stakeholders, and sustained with available resources. The following aspects demonstrate high feasibility:

User-Centric Integration

1. Ease of Use: The system will provide a Human-in-the-Loop Dashboard that allows data custodians and statisticians to review AI-generated recommendations before finalizing anonymization procedures.
2. Automation with Transparency: Unlike fully black-box anonymization tools, the proposed design combines explainable AI and optimization algorithms (NSGA-II) to justify why specific combinations of SDC techniques were selected. This aligns with operational practices in national statistical offices where interpretability and auditability are mandatory.
3. Low Training Overhead: Since the Agentic AI system automates the bulk of decision-making, end-users require only basic knowledge of SDC techniques to supervise and approve results, minimizing training costs.

Alignment with Existing Practices

- Compatibility with Microdata Workflows: Census bureaus and research institutions already apply a mix of traditional SDC techniques (recoding, suppression, micro aggregation) and newer methods like differential privacy. This system builds on existing workflows rather than replacing them entirely, ensuring smooth adoption.
- Incremental Deployment: The AutoSDC system can be introduced as a decision-support tool in parallel with existing anonymization pipelines, reducing operational risks.

Evidence from Literature

Several IEEE studies reinforce that AI-assisted anonymization systems are operationally feasible and effective in microdata privacy management:

1. **Nguyen et al. (IEEE Access, 2022)** – demonstrated that multi-objective optimization (like NSGA-II) in privacy-preserving data publishing significantly reduces disclosure risk while maintaining high utility, proving practical applicability.
2. **Alharbi & Takabi (IEEE TDSC, 2021)** – highlighted that hybrid approaches combining traditional SDC and AI-driven decision support enhance transparency and reduce deployment barriers in government statistical offices.
3. **Machanavajjhala et al. (IEEE ICDE, 2020)** – showed that decision-support systems for anonymization have higher adoption rates when interpretability and audit trails are included.

These works confirm that the proposed Agentic AI-based AutoSDC system is not only technically sound but also **operationally viable in real-world statistical environments**.

3.1.3 Legal and Ethical Feasibility

The proposed research project, conducted in collaboration with the Department of Census and Statistics of Sri Lanka (DCS), requires strict adherence to both legal and ethical standards due to its direct involvement with sensitive census microdata. Census datasets inherently contain personal and household-level information, making data protection, privacy preservation, and ethical handling critical throughout the project lifecycle.

Legal Feasibility

From a legal standpoint, the project is designed to comply with national and international data protection frameworks. In the Sri Lankan context, the Data Protection Act No. 9 of 2022 governs the use, processing, and sharing of personally identifiable information (PII). The research methodology will ensure full compliance with this regulation by implementing statistical disclosure control (SDC) techniques and modern anonymization practices, thereby preventing any possibility of re-identification of individuals. Additionally, international guidelines such as the United Nations Fundamental Principles of Official Statistics emphasize confidentiality and responsible dissemination of statistical outputs, which this project strictly follows [1].

Furthermore, ethical alignment with OECD recommendations on data governance and best practices for statistical confidentiality ensures that the project's outcomes can be trusted for responsible policy-making and research use. Legal feasibility is further strengthened by the controlled partnership with the DCS, which grants legitimate access to anonymized census datasets under institutional data-sharing agreements.

Ethical Feasibility

Ethically, the project is designed with a strong emphasis on transparency, accountability, and fairness. The use of Agentic AI for recommending combined SDC techniques and automating the anonymization process introduces new challenges in terms of algorithmic fairness and bias control. To address this, the system design ensures that AI recommendations will always prioritize privacy-preservation, non-maleficence, and equitable data utility across population subgroups [2].

In alignment with ethical standards of statistical research, no raw personally identifiable data will ever be exposed. Ethical review procedures will be conducted under the guidance of the DCS and academic supervisory committees, ensuring transparency and integrity in execution.

3.2 Requirement Gathering

3.2.1 Institutional Requirements

- **Confidentiality and Authorization:**
  The detailed anonymization methods, SDC rules, and dataset characteristics required for system design will only be provided once the official permission letter from SLIIT is issued. Until then, development will focus on generic, standard-compliant anonymization scenarios.

- **Compliance with Guidelines:**
  The system must strictly follow the anonymization and data utility guidelines set forth by the institution and any governing data protection frameworks (e.g., GDPR-like principles, census bureau practices).

3.2.2 Learning and Student Requirements

1. Knowledge Acquisition: Students need to acquire strong domain knowledge in *Statistical Disclosure Control*, which is a niche and evolving field. This represents a major learning curve.

2. Framework Familiarization: Students must learn Agentic AI frameworks (e.g., LangChain, CrewAI) and integrate them into real-world workflows.

3. Optimization Algorithms: Understanding and implementing NSGA-II for multi-objective optimization is required.

4. Collaboration Skills: Students will need to coordinate with supervisors, domain experts, and possibly external anonymization specialists for requirement clarification.

5. Research & Development: Given the novelty of integrating Agentic AI with SDC, students will spend significant time experimenting, prototyping, and validating approaches.

3.2.3 Additional Requirements

1. **Dataset Requirements**: Census microdata will be the primary dataset, provided only after SLIIT permission letter clearance. For development and testing, we still have 25% of the dataset provided by the Census department and will be used to simulate anonymization scenarios.
2. **Documentation Requirements:** Comprehensive system documentation, including design, architecture, anonymization methods, and user manuals, must be produced.
3. **Evaluation Requirements:** The system will be evaluated based on privacy risk metrics (e.g., re-identification probability) and utility metrics (e.g., information loss, statistical validity).

3.2.4 Hardware & Software Requirements

Baseline: Intel i7 or Ryzen 7 CPU, 16GB RAM, 1TB SSD, Ubuntu 22.04/Windows 11. Software: Python 3.10+, LangChain, CrewAI, DEAP/Pymoo, Pandas, TensorFlow/PyTorch, PostgreSQL/MongoDB/ Vector, Docker, Kubernetes.

3.3 Software and System Approach

3.3.1 System Overview

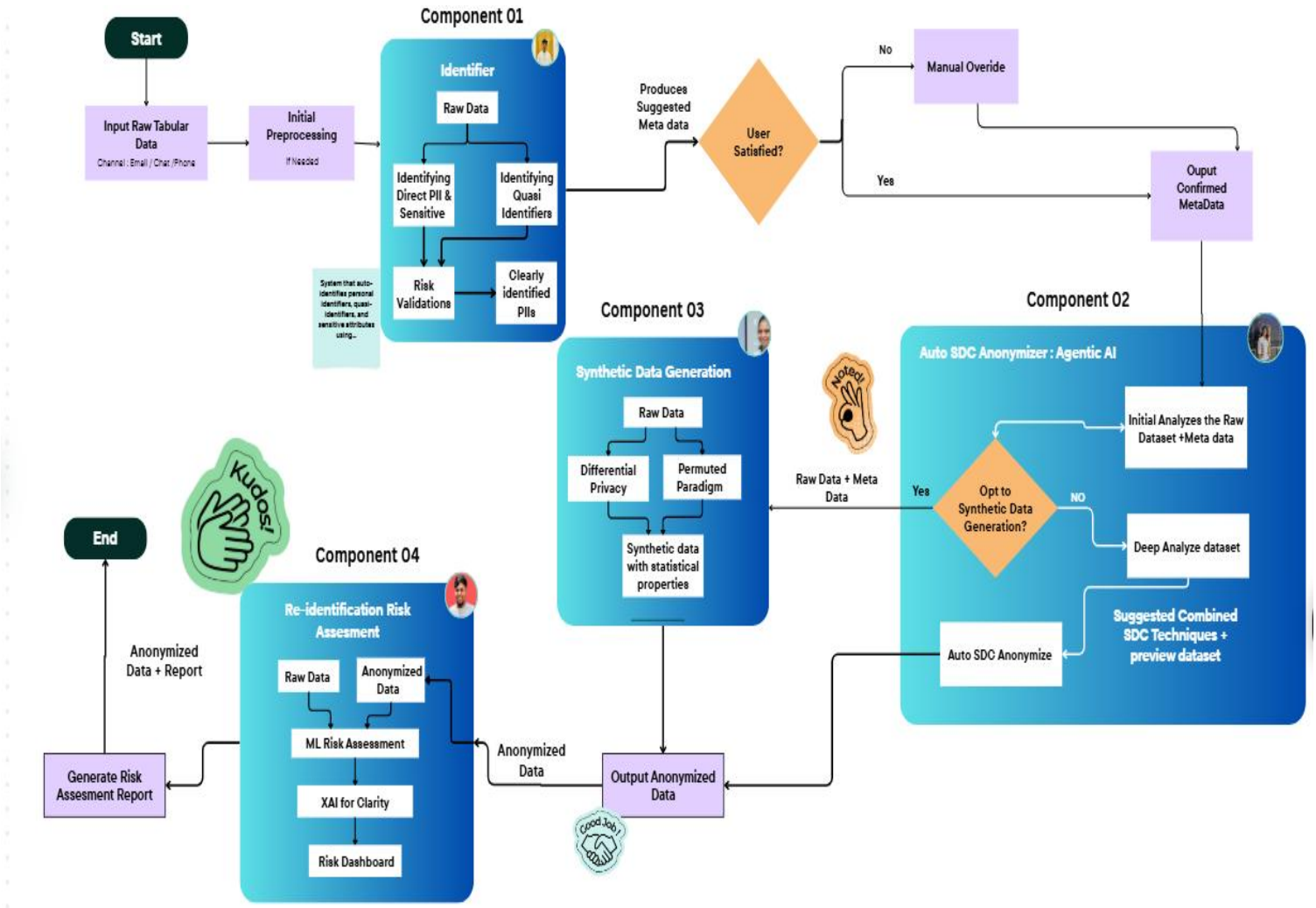

Figure 2 : Overview of the Proposed AI-Driven SDC System Components

The proposed system is an **Agentic AI–driven Statistical Disclosure Control (SDC) system** that automatically recommends, implements, and optimizes anonymization strategies for Sri Lankan census microdata. The system integrates three main modules handled by different project members:

1. **Data Profiling & Attribute Classification (Member 01):** Identifies Personally Identifiable Information (PII), Quasi-Identifiers (QIs), and Sensitive Attributes using domain knowledge and metadata.
2. **Agentic AI Recommender & Auto-Anonymization (Member 02):** Uses a rule-based knowledge system to select and implement SDC methods, auto-tunes parameters, and provides explainable justifications.
3. **Synthetic Data Generator (Member 03):** Produces synthetic datasets when traditional anonymization cannot meet risk–utility thresholds.

Finally, a **Risk & Re-identification Analysis Module** evaluates disclosure risk and utility preservation, feeding results back into the pipeline for refinement.

3.3.2 High-Level System Architecture of the Individual Component

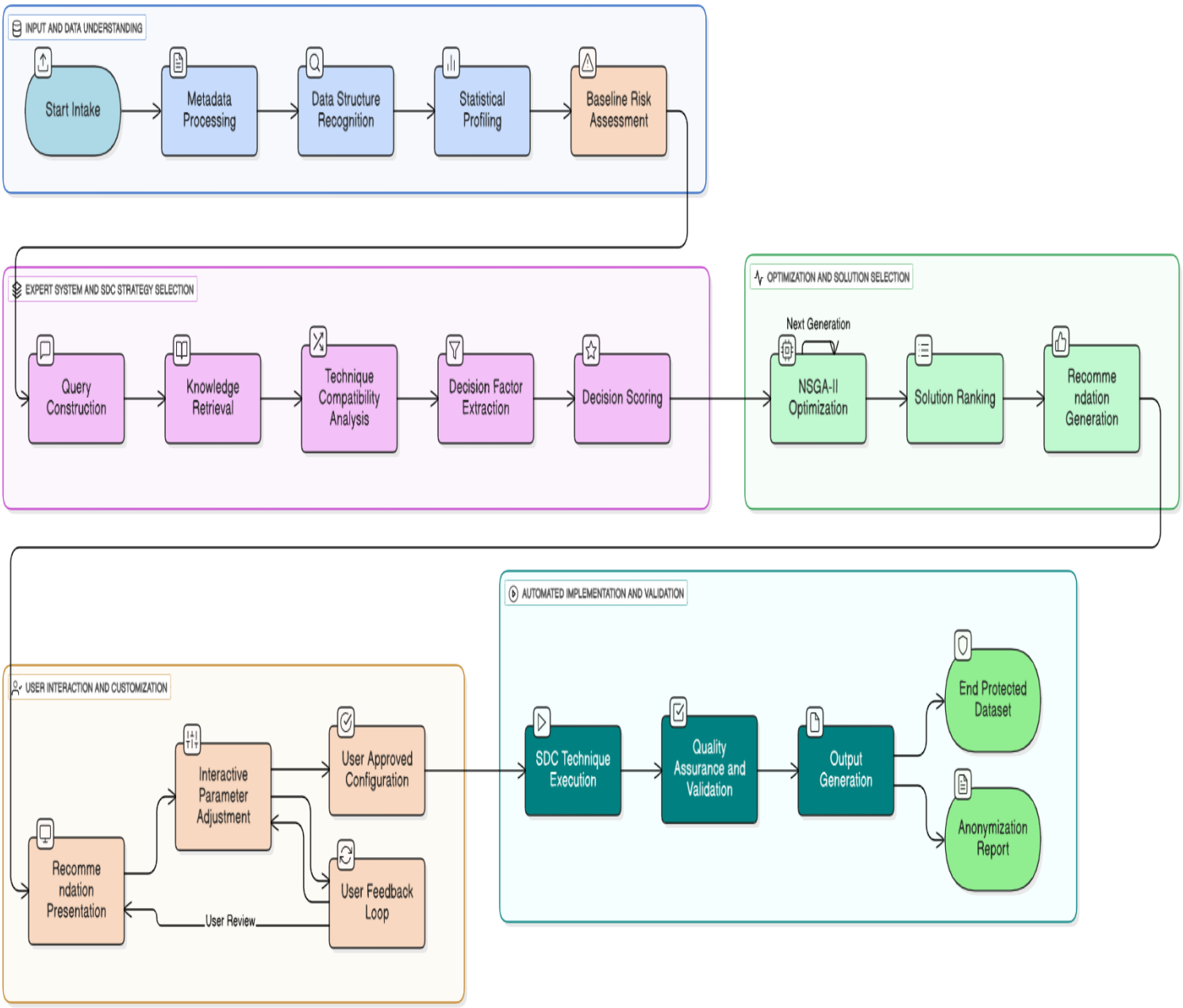

Figure 3 : High Level Architecture of the Individual Sub Component

The above figure 4 illustrates a methodology that follows a **mixed-methods experimental design** combining quantitative performance evaluation with qualitative expert validation over a 24-month research timeline. The **theoretical foundation** builds upon the established privacy-utility trade-off framework in statistical disclosure control, extending it through agentic AI principles of autonomous decision-making, goal-oriented behavior.

System Architecture and Component Development

1. **Input and Data Understanding Module**

The un- Anonymized Data is given along with meta data to the system.
The **statistical profiling components** form the foundation of the agentic system. The methodology involves developing automated data characterization algorithms that analyze:

- ✓ **Statistical profiling**: Computing distribution parameters, correlation matrices, and unique value frequencies to assess re-identification risks using statistical computation engines.
- ✓ **Risk assessment baseline**: Implementing multiple risk metrics including k-anonymity violations, l-diversity measures, and SUDA scores for comprehensive disclosure risk evaluation.

The data understanding module employs pattern matching and rule-based parsing techniques to analyze metadata descriptions and automatically categorize variables according to their sensitivity levels and disclosure potential.

2. **Expert System and SDC Strategy Selection**

The knowledge retrieval and decision-making framework represents the core intelligence of the agentic system. The methodology incorporates:

Knowledge Base Construction: A comprehensive repository of SDC methods, their effectiveness parameters, and applicability conditions will be developed through systematic literature review and expert consultation. This includes:

- ✓ Perturbative techniques (noise addition, PRAM, micro aggregation) with effectiveness parameters
- ✓ Non-perturbative methods (recoding, suppression, sampling) with utility preservation characteristics
- ✓ Hybrid approaches combining multiple techniques for enhanced protection.

Technique Compatibility Analysis: Rule-based expert systems will assess method compatibility based on data characteristics, regulatory requirements, and user specifications. The system employs multi-criteria decision analysis and logic-based reasoning to rank SDC combinations according to:

- ✓ Technical feasibility given data structure constrains.
- ✓ Regulatory compliance with data protection standards
- ✓ Computational efficiency and scalability requirements

Decision Factor Extraction: The agentic system employs statistical analysis and heuristic algorithms to identify critical decision factors influencing SDC method selection. This includes automated extraction of:

- ✓ Data sensitivity patterns and risk hotspots through statistical threshold analysis
- ✓ User utility requirements and analytical objectives via requirement parsing.
- ✓ Performance constraints and computational budgets through resource profiling

3. Multi-Objective Optimization Framework

The NSGA-II implementation forms the optimization core for balancing privacy and utility objectives. The methodology develops:

Objective Function Formulation: Mathematical formulations for competing objectives:

- ✓ Privacy objective: Composite risk measure incorporating multiple disclosure risk metrics (P = f(k-anonymity, l-diversity, t-closeness, SUDA scores))
- ✓ Utility objective: Multi-dimensional utility preservation measure (U = g(statistical accuracy, analytical validity, information preservation))
- ✓ Efficiency objective: Computational resource optimization (E = h(processing time, memory usage, implementation complexity))

Pareto Front Generation: The NSGA-II algorithm generates non-dominated solutions representing optimal privacy-utility trade-offs. The methodology includes:

- ✓ Population initialization: Diverse SDC method combinations as initial candidate solutions
- ✓ Fitness evaluation: multi-objective assessment using privacy, utility, and efficiency metrics.
- ✓ Selection and reproduction: Elite preservation and genetic operators for solution evolution

Solution Ranking: Automated ranking algorithms evaluate Pareto optimal solutions using:

- ✓ Hypervolume indicator: Measuring solution set quality and diversity.
- ✓ Inverted Generational Distance: Assessing convergence to true Pareto front.
- ✓ User preference integration: Incorporating stakeholder priorities through weighted scalarization and preference elicitation.

4. **Human-in-the-Loop Interface**

- ✓ Graphical User Interface (GUI) designed for **non-technical users**.
- ✓ Provides explanations of chosen methods, editable options, and "override" capability where humans can adjust anonymization rules manually.
- ✓ Ensures shared control: AI provides automation, humans apply contextual judgment.

Experimental Design and Validation

Evaluation datasets: synthetic, public benchmarks, and census microdata (via DCS). Metrics: Privacy (k-anonymity, re-id probability, l-diversity, SUDA), Utility (info loss, statistical accuracy), and Efficiency (processing time, memory). Baseline tools: ARX, sdcMicro, Mu-ARGUS.

3.4.3 Justification of Architecture

- User-centric design: Non-technical stakeholders can operate the system due to the human-in-the-loop and GUI-driven workflow.
- Adaptability: AI-driven recommendations dynamically adapt to different datasets without requiring custom coding.
- Transparency & Trust: The explainability module justifies anonymization choices, improving user confidence.
- Scalability: Modular structure allows adding new SDC methods (e.g., synthetic data generators) with minimal refactoring.
- Compliance: Built-in privacy rules ensure datasets align with international regulations.

Self-Evaluation Plan

Table 5 : System Component Evaluation (Weeks 1-2)

Component	Evaluation Criteria	Success Metrics	Validation Method	Timeline
Data Intake & Processing	- Handles various data formats -Accurate metadata extraction - Correct PII/QI identification	- 95% accuracy in PII detection - Support for 5+ data formats - Processing time < 2min for 10K records	- Test with DCS sample datasets - Benchmark against manual identification	Week 1
Statistical Profiling	- Comprehensive data profiling - Risk assessment accuracy - Performance on large datasets	- Complete profile generation - Risk scores align with expert assessment - Handles datasets up to 100K records	- Compare with R/Python profiling tools DCS Staff validation sessions	Week 1-2
Expert Knowledge System	- Coverage of SDC techniques - Rule accuracy - Technique compatibility logic	- 90% technique recommendation accuracy - Coverage of 15+ SDC methods - Zero incompatible recommendations	- Cross-validation with SDC literature - DCS Staff panel review	Week 2
Decision Support System	- Query construction accuracy - Knowledge retrieval effectiveness - Decision factor extraction	- 85% relevant technique retrieval - Complete factor identification - Consistent scoring across similar datasets	- A/B testing with different datasets - Expert evaluation sessions	Week 3

NSGA-II Optimization	- Convergence performance - Pareto front quality - Parameter sensitivity	- Convergence within 50 generations - Clear trade-off visualization - Stable results across runs	- Mathematical validation - Sensitivity analysis - Benchmark against other MOO algorithms	Week 3-4
Anonymizatio n Quality	- Zero re-identification risk - Preservation of data structure	- 100% anonymization - k-anonymity ≥ 3 - l-diversity maintained where applicable	- Re-identification attack simulation - Privacy metric calculation - Expert privacy assessment -Comparing results of other Anonymization apps	Week 5
Data Utility Preservation	- Statistical property retention - Analysis result consistency - Information loss minimization	- x% information loss (Threshold will be recommended by DCS) - Key statistics preserved by checking using manual programming - Correlation structures maintained	DCS Staff Review	Week 5-6
DCS Requirements Compliance	- Non-technical user usability Technique editability -Complete anonymization - Utility balance	- 90% of task completion by DCS staff - Successful parameter modification - Zero privacy breaches< - Acceptable utility retention	- DCS staff pilot testing - Requirement checklist validation - Compliance audit	Week 7

Risk Mitigation and Contingency Planning

A proactive risk analysis was conducted to identify potential challenges to the project's success. The mitigation strategies for the highest-priority risks are detailed in Table 6.

Table 6 : Risk Mitigation Strategies for High-Priority Project Risks

Risk	Probability	Impact	Mitigation Strategy
Expert knowledge incompleteness	Medium	High	Continuous literature review, expert consultations
NSGA-II convergence issues	Low	Medium	Implement alternative optimization algorithms
User interface complexity	Medium	High	Iterative design with user feedback
Performance bottlenecks	Medium	Medium	Profiling and optimization, parallel processing

Success Criteria Summary

Minimum Viable Product (MVP) Requirements:

- ✓ 85% appropriate SDC technique recommendation
- ✓ Complete anonymization with < X % utility loss (Threshold recommended by DCS)
- ✓ Usable by non-technical DCS staff
- ✓ Processing capability for typical DCS dataset sizes

Critical Success Factors:

- ✓ Early and continuous stakeholder engagement with DCS
- ✓ Iterative development with frequent validation checkpoints
- ✓ Focus on core functionality before advanced features.
- ✓ Parallel development of knowledge base and system components

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4. PROJECT REQUIREMENTS

4. 1. Functional Requirements

1. **Data Format Support**: The system shall accept and process census microdata in multiple formats including CSV, Excel, SPSS, SAS, and SQL database exports, with automatic format detection and validation.

2. **SDC Method Selection**: The system shall employ an agentic AI engine to recommend appropriate combinations of SDC techniques (global recoding, local suppression, micro aggregation, perturbation, PRAM, noise addition, synthetic data generation) based on dataset characteristics and user requirements.

3. **Parameter Optimization**: The system shall automatically tune SDC method parameters using NSGA-II multi-objective optimization to achieve optimal privacy-utility trade-offs without manual intervention.

4. **Multi-Technique Application**: The system shall apply selected SDC techniques individually or in combination, maintaining compatibility and preventing conflicting transformations.

5. **Quality Assurance**: The system shall validate anonymization results against privacy requirements and utility preservation thresholds, triggering refinement iterations when necessary.

6. **Interactive Dashboard**: The system shall provide a web-based graphical user interface allowing non-technical users to review AI recommendations, modify parameters, and approve anonymization strategies.

7. **Explainable AI**: The system shall generate clear, non-technical explanations for all anonymization decisions, method selections, and parameter choices to ensure user understanding and trust.

8. **Manual Override Capability**: The system shall allow users to override AI recommendations, adjust anonymization parameters, and apply custom anonymization rules while maintaining system guidance.

4.2. Non-Functional Requirements

- ✓ **User Experience**: The system shall achieve a System Usability Scale (SUS) score of at least 85% when evaluated by non-technical DCS staff members.
- ✓ **Learning Curve**: The system shall enable new users to complete basic anonymization tasks within 2 hours of initial training.
- ✓ **Accessibility**: The system shall comply with web accessibility standards and provide clear error messages and guidance throughout the user workflow.
- ✓ **Regulatory Compliance**: The system shall adhere to Sri Lanka Data Protection Act No. 9 of 2022, GDPR principles, and international statistical confidentiality standards.
- ✓ **Reproducibility**: The system shall ensure that identical input datasets and parameters produce identical anonymized outputs, with complete documentation of all transformation steps.
- ✓ **Quality Standards**: The system shall meet established statistical disclosure control standards and best practices as defined by international statistical organizations.

4.3. User Requirements

4.3.1 Statistical Officers (Primary Users)

- **Simplified Operation**: Statistical officers need an intuitive system that can perform complex anonymization tasks without requiring deep technical expertise in statistical disclosure control methods.
- **Transparent Reporting**: Users require clear, comprehensive reports explaining what anonymization techniques were applied, why they were selected, and what impact they had on data privacy and utility.
- **Compliance Assurance**: Officers need confidence that anonymized datasets meet all legal and regulatory requirements for public data release.

4.3.2 Data Scientists and Researchers (Secondary Users)

- **Utility Preservation**: Researchers require anonymized datasets that maintain sufficient analytical value for conducting meaningful statistical analysis and research studies.
- **Flexible Configuration**: Advanced users need the ability to fine-tune anonymization parameters and explore different privacy-utility trade-off scenarios.
- **Validation Tools**: Researchers require tools to assess the impact of anonymization on their specific analytical objectives and research questions.

User Expectations

- ✓ Minimal technical expertise required to operate.
- ✓ Ability to manually override AI recommendations.
- ✓ Access to visualizations of privacy–utility trade-offs.
- ✓ Assurance that anonymized datasets cannot be re-identified.

Note: Other necessary requirements from DCS are not revealed yet until SLIIT PERMISSION LETTER is produced.

Refer Appendix A for SRS (System Requirement Specification)

5. RESEARCH WBS AND TIMELINE

A detailed work breakdown structure and timeline for the project is provided in Figure 7 and 6, highlighting major tasks such as knowledge base construction, system development, testing, and project management activities.

5.1 Work Breakdown Structure

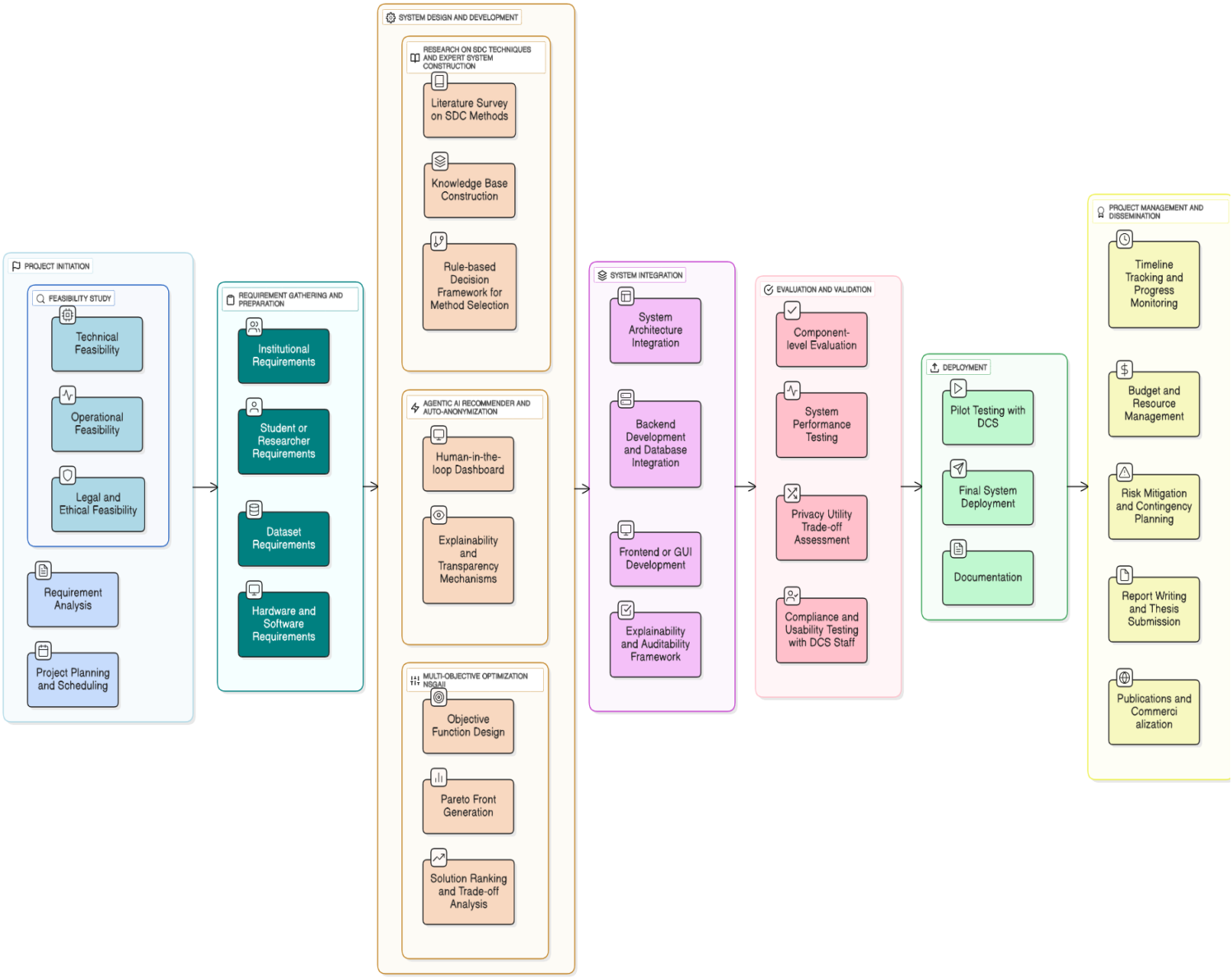

Figure 4 : Research Project Work Breakdown Structure for Building AUTOSDC

5.2 Gantt Chart

	Activity	Start	End
1	Phase 1: Research Topic Finalization	29-07-25	19-09-25
1.1	TAF Submission and Requirement Gathering	29-07-25	31-08-25
1.2	Proposal Presentation	01-09-25	08-09-25
1.3	Final Research Proposal	08-09-25	19-09-25
2	Phase 1: Foundation & Research Intensificati...	22-09-25	05-10-25
2.1	Intensive SDC Technical Deep-Dive	22-09-25	03-10-25
2.2	DCS Site Visit & Requirements Gathering	23-09-25	25-09-25
2.3	Document SDC technique comparison matrix	23-09-25	03-10-25
2.4	Develop preliminary SDC Technique classific...	25-09-25	01-10-25
2.5	Refining Raw Dataset and Anonymized dataset	24-09-25	05-10-25
3	Phase 2 : Working Prototype Development	06-10-25	03-11-25
3.1	Basic Expert System & Knowledge Base Deve...	06-10-25	11-10-25
3.2	Agentic AI Frameworks Study	06-10-25	07-10-25
3.3	Multi-Objective Optimization Research	08-10-25	08-10-25
3.4	Data Processing Pipeline	09-10-25	10-10-25
3.5	Initial Create SDC technique decision trees	11-10-25	13-10-25
3.6	Agentic AI Implementation (Prototype)	14-10-25	25-10-25
3.7	Optimization Integration -NSGA -II	25-10-25	31-10-25
3.8	Testing & First Working Prototype Finalization	31-10-25	03-11-25
4	Phase 3: System Design & Core Development	04-11-25	19-12-25
4.1	DCS 2nd Visit for Protype Demonstration and...	04-11-25	09-11-25
4.2	Real SDC Knowledge Base Architecture & pro...	04-11-25	08-11-25
4.3	Research and Updating SDC Techniques in th...	04-11-25	20-11-25
4.4	Multi-Agent Architecture Setup	09-11-25	15-11-25
4.5	Well developed SDC technique recommendat...	15-11-25	17-11-25
4.6	Parameter Optimization Agent -NSGA-II	18-11-25	27-11-25
4.7	Agent Integration & Testing	27-11-25	29-11-25
4.8	Finalized Working Prototype -II	29-11-25	06-12-25
4.9	DCS 3rd Visit to Demonstrate last Prototype ...	30-11-25	05-12-25
4.10	Presentation preparation and Presentation W...	07-12-25	19-12-25
5	Phase 4 : Application & UI Foundation	28-12-25	31-01-26
5.1	Post Progress Presentation Component Adju...	28-12-25	06-01-26
5.2	Basic Compatible Frontend & Backend Frame...	03-01-26	06-01-26
5.3	Feb 8-15: Basic UI Implementation with Worki...	07-01-26	10-01-26
5.4	Design Basic user interface mockups and wo...	11-01-26	12-01-26
5.5	Create SDC recommendation display panels ...	13-01-26	28-01-26
5.6	DCS 4th Visit for UI development Feedback a...	16-01-26	20-01-26
5.7	Build user preference and configuration man...	21-01-26	24-01-26
5.8	Create and Finalized Fully Developed Local W...	25-01-26	31-01-26
6	Phase 5 : Application Development & Testing	01-02-26	14-03-26
6.1	Basic Main Testing & Validation	01-02-26	05-03-26
6.2	Complete Additional Features & Integration o...	06-02-26	11-03-26
6.3	Finalized Testing and Validation	12-02-26	15-02-26
6.4	Backup period for Neccessary Adjustments/ ...	15-02-26	28-02-26
6.5	Presentation preparation and Presentation - I...	01-03-26	14-03-26
7	Phase 6 : System Finalization Documentation	15-03-26	30-04-26
7.1	Post Progress Presentation Component Adju...	15-03-26	31-03-26
7.2	DCS 5th visit for Final Feedbacks	25-03-26	31-03-26
7.3	Preparing Final Reports and Submission	22-03-26	15-04-26
7.4	Backup period - II for Neccessary Adjustmen...	15-04-26	30-04-26
8	Phase 7 : & Academic Completion	01-05-26	17-06-26
8.1	Final System Presentation and Viva Preparati...	01-05-26	15-05-26
8.2	Research Documentation	16-05-26	31-05-26
8.3	DCS System Handover Final	01-06-26	17-06-26

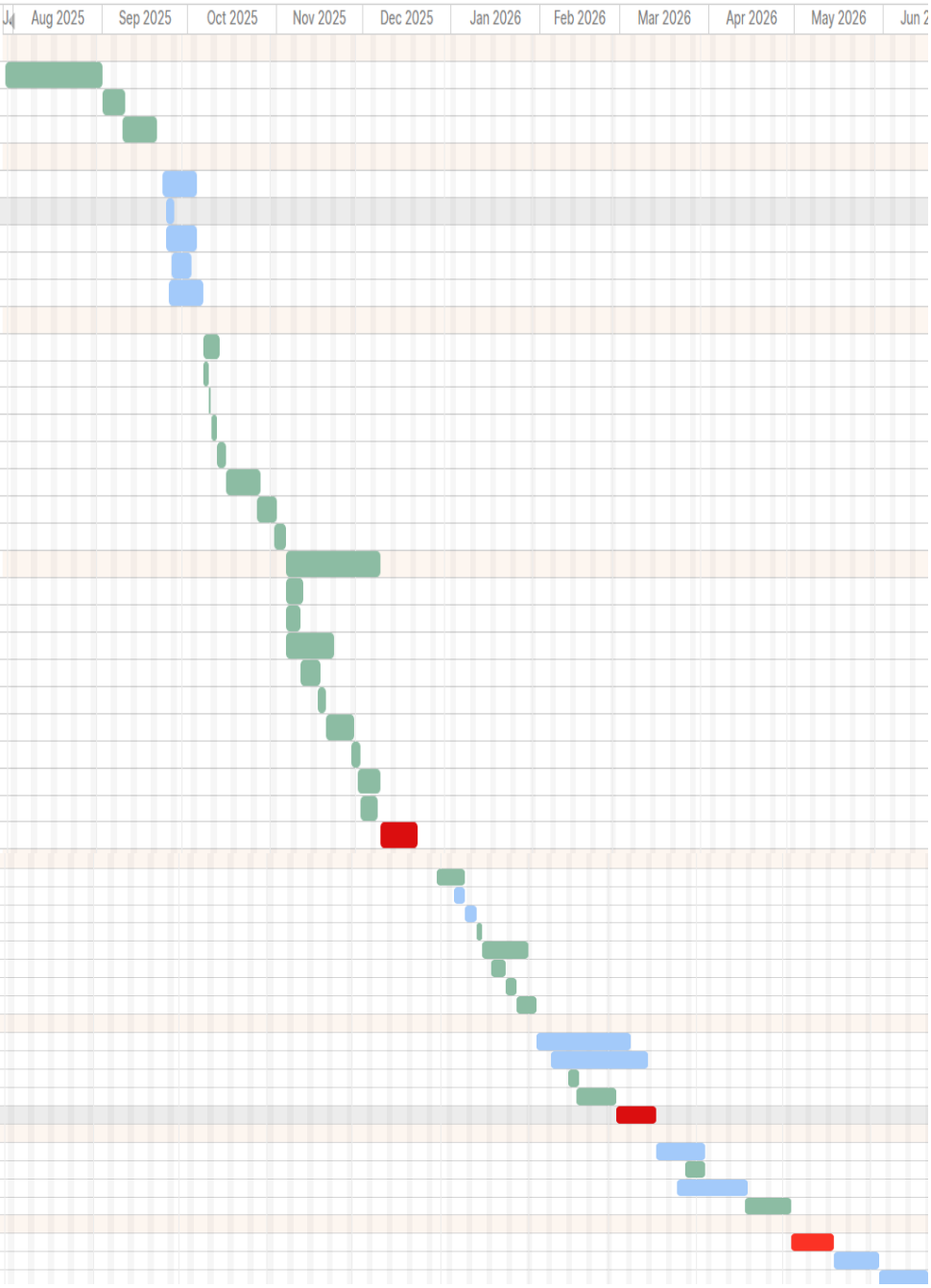

Figure 5 : Research Project Gantt Chart for Building AUTOSDC

6. BUDGET ALLOCATION

A detailed breakdown of the estimated project costs is presented in Table 7. The budget has been allocated across key categories including resources, travel, and dissemination to ensure the successful completion of all research objectives.

Table 7 : Draft Budget Allocation for the Research Project

Category	Item	Quantity	Unit Cost (Rs.)	Total (Rs.)
Hardware & Infrastructure	Laptop / Workstation (Intel i7, 16GB RAM, 1TB SSD)	1	Already Have	0
Subtotal				**0**
Software & Tools	Python & R Libraries (open source)	-	Free	0
	Development Tools (VS Code, Jupyter, GitHub)	-	Free	0
	Docker & Kubernetes (open source)	-	Free	0
Resources	Research Assistant LLMs	1	6,000	54,000
	Domain Expert Journals that need be purchased	Approx 4	1000	4,000
	Internet Browsing	16GB data per month	1145	9160
Subtotal				**67,160**
Travel	Travel to Department of Census & Statistics (for data access & validation meetings)	10 trips	Rs. 200	2,000
	Travel to SLIIT for Supervisor meetings and physical Team Meetings.	8 trips per month	Rs.200	14,400
Subtotal				**16,400**
Dissemination & Documentation	Printing & Binding of Thesis / Reports		15,000	15,000
	Journal / Conference Publication Fees	1	12,000	12,000
	Miscellaneous (stationery, contingencies)		3000	3000
Subtotal				**30,000**

Total Estimated Budget: Rs. 113,560

7.COMMERCIALIZATION

The AutoSDC (Agentic AI for Automatic SDC Pipeline Selection) system presents strong commercialization potential due to its ability to address a pressing global need: enabling safe, compliant, and efficient release of sensitive microdata while preserving analytical value. With data privacy regulations becoming increasingly stringent worldwide, the demand for automated, explainable, and cost-effective anonymization tools is expected to grow rapidly across government, research, and industry sectors.

1. **Market Demand**

   - National statistical agencies, research organizations, and international development bodies face growing pressure to provide privacy-preserving datasets.
   - Existing tools (e.g., μ-Argus, sdcMicro) are complex, fragmented, and require expert users, leaving a gap for user-friendly AI-powered solutions.
   - The introduction of privacy regulations such as the Sri Lanka Data Protection Act (2022) and international frameworks (GDPR, HIPAA) creates regulatory-driven demand for anonymization solutions.

2. **Competitive Advantage**

   - AutoSDC integrates agentic AI, multi-objective optimization (NSGA-II), and explainable anonymization workflows, features not widely available in existing tools.
   - The human-in-the-loop design ensures trust, transparency, and usability for non-specialist staff, differentiating it from black-box anonymization systems.
   - The system is scalable and modular, making it adaptable across countries and domains beyond census data (e.g., health, education, banking).

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APPENDICES

Appendix A: SRS Documentation

System Requirements Specification (SRS)

Agentic AI System for Statistical Disclosure Control and Privacy Protection

Document Information

- **Version**: 1.0
- **Date**: September 15, 2025

1.1 Purpose

This Software Requirements Specification (SRS) document specifies the requirements for an Agentic AI System designed to automate Statistical Disclosure Control (SDC) processes for microdata anonymization. The system will intelligently recommend and apply combined SDC techniques to protect individual privacy while optimizing data utility using multi-objective optimization algorithms.

1.2 Intended Audience

This document is intended for:

- **DCS Staff**: Data Collection and Statistics personnel who will use the system.
- **SLIIT Research & Development Department**
- **Supervisor:** Dr. Prasanna Sumathipala
- **Research Team**

1.3 Product Scope

The Agentic AI System will:

- Automatically analyze microdata and metadata to assess disclosure risk
- Recommend optimal combinations of SDC techniques based on data characteristics.
- Apply selected anonymization methods with minimal human intervention.
- Balance privacy protection with data utility using NSGA-II multi-objective optimization.

2.1 Product Perspective

The Agentic AI System is a new, self-contained software solution designed specifically for statistical organizations handling sensitive microdata. The system integrates with existing data processing workflows and operates as a standalone service with web-based interfaces and API connectivity.

System Context Diagram:

- Input: Microdata files, metadata descriptions, user preferences
- Processing: Data Profiling, SDC method selection, optimization, anonymization
- Output: Protected datasets, anonymization reports

2.2 Product Features

Core Features:

- **Intelligent Data Analysis**: Automated assessment of data structure, sensitivity, and disclosure risk
- **Expert Recommendation Engine**: Rule-based system for SDC method selection and parameter tuning.
- **Multi-Objective Optimization**: NSGA-II implementation for privacy-utility trade-off optimization
- **Automated SDC Implementation**: Execution of selected anonymization techniques
- **Interactive User Interface**: Web-based dashboard for system interaction and configuration
- **Comprehensive Reporting**: Detailed documentation of anonymization processes and outcomes

User Requirements

ID	Title	Need/Description	Priority
UR-001	Data Profiling for Privacy Risk Detection	Automatically profile microdata along with its metadata which contains identified Personally Identifiable Information (PII), Quasi-Identifiers (QIIs), and highlight potential privacy disclosure risks.	High
UR-002	Intelligent Method Recommendation	Receive expert-level recommendations for SDC methods based on data characteristics	High
UR-003	One-Click Anonymization	Apply recommended SDC methods with minimal manual intervention	High
UR-004	Advanced Optimization Control	Fine-tune privacy-utility trade-offs for specific research requirements	High
UR-005	Method Combination Flexibility using Human Intervention	Apply sophisticated combinations of multiple SDC techniques as Users wish	Medium
UR-006	Analytical Validation	Verify that anonymized data maintains analytical validity for research purposes	High
UR-007	Compliance Verification	Ensure anonymization meets legal and regulatory requirements	High

Functional Requirements

ID	Title	Need/Description	Priority
UR-012	Performance Monitoring	Monitor system performance and resource utilization	Low
FR-001	Data Ingestion and Format Support	Accept and parse multiple microdata formats Inputs: Data files in CSV, SPSS, SAS, Excel, JSON, XML formats; metadata descriptions Processing: Automatic format detection, schema extraction, data type inference Outputs: Standardized internal data representation, parsing status report	High
FR-002	Metadata Processing and Schema Analysis	Extract and analyze dataset structure and variable characteristics Inputs: Dataset schema, variable descriptions, data dictionary Processing: Variable classification (identifier, quasi-identifier, sensitive, non-sensitive), relationship detection Outputs: Structured metadata, variable classification report	High
FR-003	Statistical Profiling and Characterization	Compute comprehensive statistical profiles of datasets Inputs: Parsed dataset, variable classifications Processing: Descriptive statistics, distribution analysis, correlation computation, uniqueness assessment Outputs: Statistical profile report, data quality metrics 4.2 Expert System and Strategy Selection	High
FR-005	Knowledge Base Management	Maintain comprehensive repository of SDC methods and expertise Inputs: SDC method specifications, effectiveness parameters, applicability rules Processing: Knowledge base	High

		updates, rule validation, consistency checking Outputs: Current knowledge base status, method availability report	
FR-007	Technique Compatibility Analysis	Assess compatibility and effectiveness of SDC method combinations Inputs: Available SDC methods, data constraints, performance requirements Processing: Compatibility matrix evaluation, performance prediction, constraint checking Outputs: Compatibility assessment, method combination recommendations	High
FR-008	Decision Factor Extraction and Scoring	Extract and rank critical factors influencing SDC decisions Inputs: Data characteristics, user preferences, regulatory requirements Processing: Factor importance analysis, scoring algorithm application, ranking computation Outputs: Decision factor rankings, importance scores, recommendation rationale 4.3 Optimization and Solution Selection	Medium
FR-009	NSGA-II Multi-Objective Optimization	Optimize privacy-utility trade-offs using evolutionary algorithm Inputs: SDC method combinations, objective functions (privacy, utility, efficiency) Processing: Population initialization, fitness evaluation, selection, crossover, mutation Outputs: Pareto optimal solutions, convergence metrics, optimization report	High
FR-011	Interactive Preference Elicitation	Capture and integrate user preferences for solution selection Inputs: User interactions, preference specifications, constraint definitions Processing: Preference modeling, weight assignment, constraint validation Outputs: User preference profile,	Medium

		weighted evaluation criteria 4.4 User Interaction and Customization	
FR-012	Web-Based User Interface	Provide comprehensive web interface for system interaction Inputs: User actions, system status, data visualizations Processing: User interface rendering, interaction handling, session management Outputs: Interactive web pages, user feedback, session data	High
FR-013	Interactive Parameter Adjustment	Enable real-time adjustment of anonymization parameters Inputs: Parameter modifications, constraint specifications, user preferences Processing: Parameter validation, impact computation, constraint checking Outputs: Updated parameters, impact visualization, validation results 4.5 Automated Implementation and Validation	Medium
FR-015	SDC Technique Execution Engine	Execute selected SDC methods on target datasets Inputs: Dataset, selected SDC methods, method parameters Processing: Method execution, progress monitoring, error handling Outputs: Anonymized dataset, execution log, performance metrics	High
FR-016	Quality Assurance and Validation	Validate anonymization quality and compliance Inputs: Anonymized dataset, original dataset, quality criteria Processing: Privacy verification, utility assessment, compliance checking Outputs: Validation report, quality metrics, compliance status	High

on-Functional Requirements

ID	Title	Need/Description	Priority
FR-017	Output Generation and Export	Generate and export anonymized datasets in required formats Inputs: Anonymized dataset, export format specifications, metadata Processing: Format conversion, metadata preservation, file generation Outputs: Exported files, export status, format validation 5. Non-Functional Requirements 5.1 Performance Requirements	High
NFR-004	Memory Efficiency	Process datasets using <4GB RAM per dataset up to 1 million records	Medium
NFR-006	Error Recovery	Automatic recovery from transient failures within 5 minutes	Medium
NFR-009	Learning Curve	New users achieve task completion within 2 hours of training	Medium
NFR-010	User Satisfaction	Achieve >4.0/5.0 average user satisfaction rating	Medium
NFR-015	Browser Compatibility	Support latest versions of Chrome, Firefox, Safari, and Edge browsers	Medium

External Interface Requirements

ID	Title	Need/Description
NFR-017	Data Format Support	Native support for CSV, SPSS, SAS, Excel formats with 99%+ parsing accuracy
UI-001	Primary Web Interface	Responsive web application providing complete system functionality Components: Dashboard, data upload, method selection, parameter configuration, results visualization Design Standards: Material Design principles, accessibility compliance Screen Resolution: Support 1024x768 to 4K displays Browser Requirements: Modern browsers with JavaScript enabled 6.2 Hardware Interfaces
HW-001	Server Hardware	• CPU: 16 cores, 3.0GHz • RAM: 64GB • Storage: 1TB NVMe SSD • Network: 10 Gigabit Ethernet
HW-002	Client Hardware	• CPU: Dual-core 2.0GHz • RAM: 4GB • Display: 1024x768 resolution • Network: Broadband internet connection • **Input Devices**: Standard keyboard and mouse, touch screen support
COM-001	Network Protocols	• HTTP/HTTPS: Web interface communication, API endpoints

		• TLS 1.3: Encrypted communication for all external interfaces • TCP/IP: Standard network communication protocols • WebSocket: Real-time updates and progress monitoring
COM-002	Data Exchange Formats	• Input Formats: CSV, TSV, SPSS (sav), SAS (. sas7bdat), Excel (xlsx), JSON, XML • Output Formats: Same as input plus anonymized variants. • Metadata Formats: JSON schema, XML schema, custom metadata formats • Report Formats: PDF, HTML, CSV summary reports

Assumptions and Dependencies

ID	Title	Need/Description
TECH-003	Integration Constraints	
ASMP-001	User Expertise	• Assumption: Users have basic statistical knowledge and data handling experience • Rationale: Target users are DCS professionals with domain expertise • Impact: User interface complexity, help system requirements • Validation: User surveys, competency assessments during requirements gathering
ASMP-002	Data Quality	• Assumption: Input data sets meet basic quality standards with proper metadata • Rationale: Professional statistical organizations maintain data quality standards • Impact: Validation requirements, error handling complexity • Validation: Sample dataset analysis, quality assessment procedures
ASMP-003	Technology Environment	• Assumption: Target deployment environments have modern IT infrastructure • Rationale: Statistical organizations typically maintain current technological systems • Impact: Technology selection, performance requirements, deployment approach
DEP-001	Third-Party Software	○ Python ecosystem libraries (NumPy, Pandas, Scikit-learn, R Programming)

		○ Database management systems (PostgreSQL, MySQL, Vector database) ○ Web development frameworks (Django/Flask, React/Vue.js) • **Risk**: License changes, version compatibility issues, security vulnerabilities • **Mitigation**: Version pinning, license monitoring, security update procedures
DEP-002	Statistical Expertise	• **Dependencies**: ○ Domain expert input for knowledge base construction ○ Validation from statistical disclosure control specialists ○ Regulatory guidance from privacy officers • **Risk**: Limited expert availability, conflicting expert opinions • **Mitigation**: Multiple expert consultations, structured knowledge elicitation, conflict resolution procedures

Appendix B: Requirements Traceability Matrix

Requirement ID	Priority	User Story	Functional Requirement	Test Case
UR-001	High	As a data analyst, I need automated risk assessment	FR-004	TC-001
UR-002	High	As a data analyst, I need intelligent recommendations	FR-007, FR-008	TC-002
UR-003	High	As a data analyst, I need one-click anonymization	FR-015	TC-003
UR-004	High	As a researcher, I need optimization control	FR-009, FR-011	TC-004
UR-005	Medium	As a researcher, I need method combinations	FR-007	TC-005
UR-006	High	As a researcher, I need analytical validation	FR-016	TC-006

Appendix C: Risk Assessment Matrix

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
RSK-001	SDC method failures	Medium	High	Comprehensive testing, fallback methods
RSK-002	Performance degradation	Medium	Medium	Load testing, optimization procedures
RSK-003	Compliance violations	Low	High	Legal review, compliance testing
RSK-004	Data security breaches	Low	High	Security audits, encryption protocols
RSK-005	User adoption resistance	Medium	Medium	Training programs, change management